Bow Valley College

Course Outline

SODV3201

Web Programming Foundations

Fall 2020 - Spring 2024

Last Updated: 8/31/2020 2:16:10 PM

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SODV3201 Web Programming Foundations

COURSE DESCRIPTION

This course focuses on foundational programming techniques underpinning web application development. Learners plan, design, program, and integrate server-side elements for web applications. Group and individual activities address common business needs such as basic security, session management, and data persistence.

REQUISITES	Complete or concurrently enroll in: • DESN3203 - Web Design and Development (3)
EQUIVALENTS	None
CREDITS	3
HOURS	45
ELIGIBLE FOR PLAR	No
ZERO TEXTBOOK COST	

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Write basic backend programs for execution on a web server.	2, 3, 4, 5, 7
2	Create server-side programs that dynamically generate web pages, clearly separating presentation and behaviour.	1, 2, 3, 4, 5, 7
3	Develop programs that validate form input on the server.	1, 2, 3, 4, 5, 7, 9
4	Design and code programs that respond to web pages and perform CRUD operations.	1, 2, 3, 4, 5, 7, 9
5	Identify issues faced by web developers such as security, file management, and web-standard compliance.	1, 2, 3, 4, 5, 6, 7, 9

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Module 1: The ASP.NET MVC Architecture • Course Introduction • Course Description and Outline • Assessment and Grading Policy • Setting Web Development Environment • Creating and Running First ASP.NET application • ASP.NET File Structure and Tools • What is inside the Solution Explorer • Learning the MVC (Model, View, and Controller)
Week 2	Module 1: The ASP.NET MVC Architecture • Programing Model-View-Controller (MVC) together • Adding a new model to project • Adding a new controller to project • Adding a new view to project • Utilizing Shared View _Layout • Introduction to Razor View Engine • How to pass a model to a view • Add a bootstrap theme to the project • Knowing Bootstrap File structure • Responsive web site design • Knowing purposes of bundles config. • install packages from Nuget Package Manager
Week 3	Module 1: The ASP.NET MVC Architecture • Learn the input/output format of an Action in a Controller • ASP.NET Route URL Pattern • Calling an action from another action in a different controller • Calling a View from an action • Call Html helper methods from an action • You will know Conventional Routing • Custom Routes • Route Constraints • Attribute Routes • Identify ways to pass data to a view • How to pass temporary data using ViewData and ViewBag • How to customize a View and reuse in the form of Partial View.

Week 4	Module 1: The ASP.NET MVC Architecture • Know the differences between Domain Models and ViewModels • Use ViewModel to transfer data from multiple domain models to a View. • The Razor View Syntax and its uses • Dynamic CSS Class Assignment using The Razor Views Module 2: The Entity Framework Code-First Approach • Object-Oriented Programming approach to Database Construction. • Know Entity Framework Code-First approach to Database Creation. • Learn about the Object Relational Mapper (ORM) • Understand (Code First Vs. Database First) • Create a database table using the Entity Framework (EF). • Learn the DbContext and DbSet • The Linq-to-Entities Query • Enabling Migrations in EF.
Week 5	Module 2: The Entity Framework Code-First Approach • Creating a New Domain Model • Updating the Model • Add-Migration and Update-Database • Understanding Package Manager Console • Populate data using Entity Framework • Overriding EF Conventions • Querying on the Database • Know Eager Loading Operation Add a new record in the database • Know The @Html Markup • The + Methods • Creating Customized Labels for Forms • Bootstrap Styling • The Dynamic Model Binding • Database Interactions • Learning Database CRUD Operations
Week 6	Module 2: The Entity Framework Code-First Approach • Edit records in Razor View • Learn about The Edit View • Updating Temporary Title Of a View • Learn Data Validations in the ASP.NET • The conditional ModelState properties • Know the Attribute Validation technique • Learn the Validation Message • Know ASP.NET Built-in Data Annotation Attributes for Validation • Apply Custom Validation
Week 7	Module 2: The Entity Framework Code-First Approach • Apply the Validation Summary • Know Server-Side-Validation • Know Client-Side-Validation • Enabling Client-Side-Validation • Limitation of Client-Side-Validation • Cross-Site Access Restriction • Prevent your application from Cross-Site Request Forgery (CSRF) Attacks Module 3: JQuery Plug-ins, Authentication, Authorization, Social-Logins & Deployment • Learn the JQuery DataTables Plug-in • Install DataTables • Optimizing Bundles and Scripts for application performance enhancement
Week 8	Module 3: JQuery Plug-ins, Authentication, Authorization, Social-Logins & Deployment • Error diagnosis and bug fixing • Learning Delete Operation on ASP.NET Applications • Know the [httpGet] and [HttpPost] attributes. • Use [HttpPost] attribute and @Html anti-forgery tokens for secure delete operation.01
Week 9	Reading Week
Week 10	Module 3: JQuery Plug-ins, Authentication, Authorization, Social-Logins & Deployment • Perform an activity (In Class) based on previous lectures • Know ASP.NET Login and Authentication • Learn ASP.NET Identity framework • Apply Access Restriction to applications • Protecting anonymous users access

Week 11	Module 3: JQuery Plug-ins, Authentication, Authorization, Social-Logins & Deployment • Understanding ASP.NET Roles (for managers, users) using ASP.NET Identity. • Setting Roles by defining RoleStore and RoleManager. • Creating RoleManager (Admin) • Learn an alternative approach to setting roles via database migrations • Modify Application Behavior according to user roles. • Create Separate Views based on Role Assignment Know the OAuth • Implement third-party authorization like social logins (Facebook, Microsoft, Twitter, or Google authentication) • Register Application with a Social Service Provider • Understand App ID and App Secret • Profile user data while new users logs-in • Use a secure SSL channel for communication
Week 12	Module 3: JQuery Plug-ins, Authentication, Authorization, Social-Logins & Deployment • Install Internet Information Server (IIS) • Learn ASP.NET application deployment to IIS on your local computer and on the web servers. • Host ASP.NET application online • Learn asynchronously technique to access web servers • Enhance responsiveness to websites using JQuery Ajax Library. • Implement the delete method using Ajax Library. • Learn to remove rows from DataTable Plugin.
Week 13	Module 3: JQuery Plug-ins, Authentication, Authorization, Social-Logins & Deployment • Delete securely and prevent CSRF using JQuery Ajax method • Install the JQuery BootBox plug-in • Use an asynchronous message dialog box from BootBox • Review class before project Presentation
Week 14	Project Presentation Demo
Week 15	Project Presentation Demo

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1, 2, 3, 4, 5	Engagement	10%
1, 2, 3, 4	Learning activities	25%
1, 2, 3, 4	Assignments	35%
1, 2, 3, 4, 5	Project	30%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A minimum grade of D is required to pass this course. However, a program may require a higher grade in this course to progress in the program or to meet specific program completion requirements.

Please consult with the program area or contact the program chair for further details. A minimum Grade Point Average of 2.0 is required for graduation.

GRADING SCHEME

Grade	Percentage	Grade Point	Description
A+	95-100	4.0	Exceptional: superior knowledge of subject matter
A	90-94	4.0	Excellent: outstanding knowledge of subject matter
A-	85-89	3.67	
B+	80-84	3.33	
B	75-79	3.0	Very Good: knowledge of subject matter generally mastered
B-	70-74	2.67	
C+	67-69	2.33	
C	64-66	2.0	Satisfactory/Acceptable: knowledge of subject matter adequately mastered
C-	60-63	1.67	
D+	57-59	1.33	
D	50-56	1.0	Minimal Pass
F	Less than 50	0.0	Fail: an unsatisfactory performance

REQUIRED LEARNING RESOURCES

Syed, B.A. (2014) Beginning node.js: Unleash the power of node.js and create highly scalable websites. (1st ed.). Berkeley, CA: Apress. ISBN: 9781484201886

Haverbeke, M. (2014) Eloquent Javascript: a Modern introduction to programming. Open Textbook Library (Corporate Author), San Francisco, CA: No Starch Press.

Http://eloquentjavascript.net/Eloquent_JavaScript.pdf"

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.